USN

[22CSDE11]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER 2023

NETWORK DEFENSE

Time: 3 Hours Maximum **Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	a.	Compare phishing and spear phishing attacks.	L-4	CO1	**02**
	b.	Explain the vulnerabilities exposed by Linux Kernel Architecture	L-3	CO2	**02**
	c.	What is meant by logical isolation?	L-1	CO3	**02**
	d.	Define Network Performance Monitoring in network operations.	L-1	CO4	**02**
	e.	List any four types of risks faced by the organization.	L-2	CO5	**02**
		PART-B	**06X 05=30**		
2.	a.	Identify common security misconfigurations that can make cloud resources vulnerable.	L-2	CO1	**05**
		OR			
	b.	Explain the social engineering tactics used in a real-world email attack and assess how they could have been prevented.	L-4	CO1	**05**
3.	a.	Describe the concept of IoT device lifecycle management and the role it plays in maintaining security. What are the key stages of an IoT device's lifecycle?	L-2	CO2	**05**
		OR			
	b.	Analyze how to enforce network perimeter security using firewalls?	L-4	CO2	**05**
4.	a.	Explain the concept of Open System Authentication in wireless networks.	L-2	CO3	**05**
		OR			
	b.	Outline the Shared Responsibility Model in cloud security, and explain the responsibilities of cloud providers and customers in detail.	L-2	CO3	**05**
5.	a.	Explain the importance of establishing baseline traffic signatures for network monitoring.	L-2	CO4	**05**
		OR			
	b.	Develop a baseline for normal traffic patterns Using a specific network scenario.	L-3	CO4	**05**
6.	a.	Explain the importance of cyber threat intelligence in predicting and preventing cybersecurity incidents.	L-2	CO5	**05**
		OR			
	b.	Design a comprehensive risk management framework tailored to the specific needs and goals of an organization.	L-3	CO5	**05**

7.	a.	Explain the importance of real-time monitoring of network logs in a cybersecurity context.	L-2	CO6	05
		OR			
	b.	Distinguish full virtualization and para-virtualization.	L-4	CO6	05
		PART-C	**06 X 10=60**		
8.	a.	Illustrate some widely used symmetric encryption algorithms and highlight their strengths and weaknesses.	L-2	CO1	10
		OR			
	b.	Explain the following with respect to Technical Network Security control i. Access control **05 M** ii. Authentication control **05 M**	L-4	CO1	10
9.	a.	Describe the key security challenges associated with mobile devices in an organization, and how do these challenges differ from traditional endpoint security?	L-2	CO2	10
		OR			
	b.	Explain the primary components of an effective endpoint security solution for Windows systems.	L-2	CO2	10
10.	a.	Draw a basic wireless network configuration for a small office environment and explain the components of a Wireless network.	L-3	CO3	10
		OR			
	b.	Write the step-by-step guide for configuring WPA encryption on a wireless access point.	L2	CO3	10
11.	a.	Apply a centralized log monitoring system to identify and troubleshoot a specific issue within a distributed network.	L-3	CO4	10
		OR			
	b.	Compare the effectiveness of Log Monitoring and Analysis on Windows Systems, Linux, and MAC Systems.	L-4	CO4	10
12.	a.	Explain the significance of threat intelligence layer to enhance the organization's cybersecurity posture.	L-2	CO5	10
		OR			
	b.	Describe the impact of the risk management phase on achieving organizational goals.	L-3	CO5	10
13.	a.	Explain a strategy to defend against a specific type of network attack, such as DDoS or ransomware.	L-3	CO6	10
		OR			
	b.	Illustrate the concept of threat modeling in administrative application security. How did it help to identify potential security risks in an application?	L-2	CO6	10